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import numpy as np
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import
accuracy_score,precision_score,recall_score,confusion_matrix,classification_report

data=pd.read_csv("winequalityN.csv")
data.isnull().sum()
data.dropna(inplace=True)

data.duplicated().sum()
data.drop_duplicates(inplace=True)

data.replace("?",pd.NA,inplace=True)
data.dropna(inplace=True)

data['type']=data['type'].map({'white':0,'red':1})
X=data[['fixed acidity','volatile acidity','citric acid','residual sugar','chlorides','free sulfur
dioxide','total sulfur dioxide','density','pH','sulphates','alcohol','quality']]
y=data['type']

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)

model=DecisionTreeClassifier()
model.fit(X_train,y_train)

#test
y_pred=model.predict(X_test)
accuracy=accuracy_score(y_test,y_pred)
precision=precision_score(y_test,y_pred)
recall=recall_score(y_test,y_pred)
f1score=f1_score(y_test,y_pred)
confusionmatrix=confusion_matrix(y_test,y_pred)
print(accuracy)
print(precision)
print(recall)
print(f1score)
print(confusionmatrix)
print(classification_report(y_test,y_pred))

import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

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plt.figure(figsize=(20,10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=["white","red"],
    filled=True,
    rounded=True
)
plt.show()
```